PHYSICS ADAPTLY

FORCES – FORCES & MOTION

KEY VOCABULARY

Resultant force: the single force that has the same effect as all forces.

Weight: the force of gravity on a mass; W = m × g.

Acceleration: the rate of change of velocity, in m/s^2.

Terminal velocity: steady speed when drag balances weight.

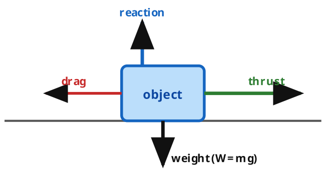
COMMON MISCONCEPTIONS

~~A moving object always has a force pushing it.~~
✓ At constant speed, forces are balanced.

~~Mass and weight are the same thing.~~
✓ Mass is in kg; weight is a force in newtons.

~~Heavier objects always fall faster.~~
✓ Without air, all objects fall at the same rate.

FORCES ON A MOVING OBJECT



If thrust = drag and reaction = weight, the resultant force is **zero**.

1 WEIGHT

A 6 kg mass sits on Earth (g = 9.8 N/kg). **Calculate** its weight. **[2 marks]**

Answer: ______________________ N

2 NEWTON'S SECOND LAW

A resultant force of 12 N acts on a 4 kg trolley. **Calculate** its acceleration (F = m a). **[2 marks]**

Answer: ______________________ m/s^2

3 RESULTANT FORCE

Explain what happens to a car's motion when the driving force equals the drag. **[2 marks]**

4 TERMINAL VELOCITY

Explain why a skydiver reaches a terminal velocity. **[3 marks]**

5 BALANCED FORCES

Complete the sentence using the word bank. **[2 marks]**

When forces are balanced the resultant force is _____ and the object moves at a _____ velocity.

zero | constant | increasing | non-zero

6 STOPPING DISTANCE

Stopping distance = thinking distance + braking distance. State **one factor** that increases each. **[2 marks]**

TIP: *Resultant force = mass × acceleration (F = m a).*

✓ **Think: Why does doubling a car's speed more than double its braking distance?**